YIFEI SUN

Software Engineer · Internal Tools, Operations Systems & Integrations · 13 Years

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SUMMARY

Full-stack engineer with 13 years building internal tools, operations systems, and third-party integrations end-to-end. Experienced across the full lifecycle: data pipelines, admin dashboards, CRM integrations, ticketing workflows, and developer tooling. Built systems from scratch at early-stage startups (Toast, first 20 engineers) through scale-up. Strong at translating messy operational problems into clean, maintainable software — partnering directly with support, ops, and business teams to ship solutions quickly. Active user of AI-assisted development tools (Claude Code, Cursor) for rapid prototyping and automation.

HIGHLIGHTS

- Built CRM integrations and partner sync systems at Toast; translated early customer feedback into platform capabilities
- Designed internal tooling and automation for engineering workflows (build/test pipelines, observability, deployment)
- Full-stack across every role: admin dashboards (Angular, TypeScript), backend services (Java, Node.js, Python), data pipelines (Kafka, SQS/SNS)
- Shipped 0→1 products under ambiguity; rapid experimentation with quick iteration based on internal and customer feedback

Technologies: Backend: Java, Node.js, Python, Kotlin, Spring Boot · **Frontend:** Angular, TypeScript, JavaScript, React (familiar) · REST API, GraphQL, gRPC · **Data:** Kafka, SQS/SNS, PostgreSQL, MySQL, Redis, Snowflake (familiar) · **Infra:** AWS, Kubernetes, Docker, Terraform · Datadog, Grafana, OpenTelemetry · CI/CD · **AI Tools:** Claude Code, Cursor, agentic workflows

INDEPENDENT PROJECT

GenAI Decision Engine & Experimentation Platform

- Architected a hybrid rules + multi-path GenAI engine (fast vs. deep inference) with a policy overlay over heterogeneous telemetry, producing explainable real-time recommendations.
- Built "Prompt Lab" for end-to-end decision simulation, offline evaluation/backtesting, and prompt/parameter search (grid & Bayesian optimization) with multi-provider LLM support.
- Designed a configurable policy framework with DB-backed configs, environment-specific overrides, and a scheduler-driven tuning loop that sweeps parameters and publishes promotable config versions.

EXPERIENCE

CarGurus — Principal Software Engineer / Technical Lead

Oct 2019 – Present

Internal Orchestration Platform & System Consolidation

- Built an orchestration service consolidating 12 downstream integrations into 5 well-defined services with circuit breaking, fault isolation, and idempotent retry — reducing operational complexity and manual triage burden.
- Designed automated parity-testing infrastructure comparing new vs. legacy system responses at 50% production traffic, enabling data-driven rollout decisions without manual QA intervention.
- Established observability standards (structured logging, distributed tracing, SLO-based dashboards) giving engineering and ops teams real-time visibility into system health across product areas.

p99: 250 ms → 50 ms (5×) **Availability: 98% → 99.9%** **Integrations consolidated: 12 → 5**

Dealer Admin Dashboard & Data Integration (GraphQL Federation)

- Built dealer-facing admin dashboards for inventory and listing management end-to-end, serving as the primary interface for CarGurus' B2B customers.
- Designed and integrated a Netflix DGS subgraph into the federated GraphQL gateway, exposing vehicle delivery data to frontend, backend, and AI consumers from a single unified graph.

Digital Commerce — Rapid Experimentation Platform

- Led backend design for a new commerce product using event-driven architecture (SQS/SNS, Kafka); built infrastructure for rapid product experimentation — launch, measure, and safely retire product bets without destabilizing core systems.
- Delivered a performant implementation under ambiguous requirements; designed for safe, incremental production rollout.

Cross-Org Standards & Developer Tooling

- Authored design docs, RFCs, and decision records adopted across 10+ teams; established patterns for API design, caching, and distributed orchestration.
- Led 10+ training sessions for engineering teams. Mentored senior engineers on system design and ownership practices.

Applause — **Senior Software Engineer / Technical Lead** Oct 2017 – Oct 2019

Internal Workflows & Enterprise Integrations

- Partnered with product, support, and customer-facing teams to modernize internal workflows and enterprise integrations — building admin tools and dashboards (Angular, TypeScript, Node.js BFF) used daily by operations staff.
- Drove design of microservice architecture based on event sourcing and CQRS; defined foundational patterns for data pipelines adopted by new domains.

Authentication Platform & API Gateway

- Upgraded legacy authentication to a modern Keycloak-based SSO platform with custom plugins (Python) for Kong API gateway. Unified auth across internal systems, enterprise partners, and external community.
- Set up API gateway and service mesh infrastructure (Kong + Linkerd on Kubernetes) with performance tuning.

Toast Inc. — **Senior Software Engineer (early team, ~20 engineers)** Sep 2014 – Oct 2017

CRM, Partner Integrations & Core Platform

- Built core backend services (Order Service, Labor Service, REST APIs) and customer-facing admin dashboards (jQuery, Angular) for restaurant owners — end-to-end ownership from database to UI.
- Led third-party integrations with key partners (workforce management, payment processors), ensuring reliable bidirectional sync. Partnered with PMs, sales, and support to translate early customer feedback into platform capabilities.
- Developed consumer-facing Android application for Toast's mobile ordering platform.

Internal Tooling & Infrastructure (Pre-DevOps)

- Set up the initial AWS environment using Ansible — provisioned infrastructure, deployment pipelines, and monitoring before a dedicated DevOps team existed.
- Built a custom API gateway in Nginx with Lua scripting for rate limiting, request routing, and traffic management.

The MathWorks Inc. — **Software Engineer II** Jul 2013 – Aug 2014

Developed internal engineering tools (Java backend) for MATLAB/Simulink workflows. Built backend automation systems for build/test reliability and developer productivity.

EDUCATION

M.S. in Electrical and Computer Engineering — Northeastern University, Boston 2010 – 2013

B.S. in Information Science — Beijing University of Aeronautics and Astronautics 2006 – 2010